Solving Linear Systems

Last Time

$$\begin{cases} 5x + y = 3 \\ 2x - y = 4 \end{cases}$$

$$\left(\begin{array}{cc|c} 5 & 1 & 3 \\ 2 & -1 & 4 \end{array}\right) \rightsquigarrow \begin{array}{c} \begin{array}{ccc} \text{"x"} & \text{"y"} & \end{array} \\ \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -2 \end{array}\right) \end{array}$$

$\Rightarrow x = 1, \; y = -2$

Elementary Row Operations:

cR_i, $c \neq 0$ $\qquad R_i \leftrightarrow R_j \qquad R_i + cR_j$, $i \neq j$

Matrices in Row-Echelon Form (REF):

① The first non-zero entry in every row is to the right of the previous one.

② Zero rows are at the bottom.

We also have Reduced REF:

③ Every leading entry must be 1, and

④ Every leading entry is alone in its column.

Ex.

	REF?	RREF?
$\left(\begin{array}{cc\|c} \boxed{1} & 3 & 4 \\ 0 & 0 & \boxed{1} \end{array}\right)$	✓	✗
$\left(\begin{array}{ccc\|c} \boxed{1} & 0 & 3 & 1 \\ 0 & \boxed{1} & 2 & 4 \end{array}\right)$	✓	✓
$\left(\begin{array}{cc\|c} \boxed{1} & 2 & 0 \\ 0 & \boxed{1} & 0 \\ 0 & 0 & 0 \end{array}\right)$	✓	✗
$\left(\begin{array}{cc\|c} \boxed{1} & 0 & 0 \\ 0 & \boxed{1} & 0 \\ 0 & \boxed{2} & 0 \end{array}\right)$	✗	✗
$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$	✓	✓

Ex. Let's solve:

$$\begin{cases} x_1 + 3x_2 - 2x_3 + 2x_5 = 0 \\ 2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 = -1 \\ 5x_3 + 10x_4 + 15x_6 = 5 \\ 2x_1 + 6x_2 + 8x_4 + 4x_5 + 18x_6 = 6 \end{cases}$$

Create the augmented matrix

$$\left(\begin{array}{cccccc|c} \boxed{1} & 3 & -2 & 0 & 2 & 0 & 0 \\ \boxed{2} & 6 & -5 & -2 & 4 & -3 & -1 \\ 0 & 0 & \boxed{5} & 10 & 0 & 15 & 5 \\ \boxed{2} & 6 & 0 & 8 & 4 & 18 & 6 \end{array}\right)$$

$\downarrow R_2 - 2R_1, \quad R_4 - 2R_1$

$$\left(\begin{array}{cccccc|c} \boxed{1} & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & \boxed{-1} & -2 & 0 & -3 & -1 \\ 0 & 0 & \boxed{5} & 10 & 0 & 15 & 5 \\ 0 & 0 & \boxed{4} & 8 & 0 & 18 & 6 \end{array}\right)$$

$\downarrow R_3 + 5R_2, \quad R_4 + 4R_2$

$$\left(\begin{array}{cccccc|c} \boxed{1} & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & \boxed{-1} & -2 & 0 & -3 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \boxed{6} & 2 \end{array}\right)$$

$\downarrow x_4 \quad x_5 \quad R_3 \leftrightarrow R_4$

$$\begin{array}{c} \begin{array}{cccccc} & x_2 & & x_4 & x_5 & \end{array} \\ \left(\begin{array}{cccccc|c} \boxed{1} & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & \boxed{-1} & -2 & 0 & -3 & -1 \\ 0 & 0 & 0 & 0 & 0 & \boxed{6} & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \end{array}$$

REF ✓

Thus:

$$\begin{cases} x_1 + 3x_2 - 2x_3 + 2x_5 = 0 & (1) \\ -x_3 - 2x_4 - 3x_6 = -1 & (2) \\ 6x_6 = 2 & (3) \\ 0 = 0 \end{cases}$$

(3): $x_6 = \frac{2}{6} = \frac{1}{3}$

(2): $-x_3 = -1 + 2x_4 + 3x_6$

$x_3 = 1 - 2x_4 - 3\left(\frac{1}{3}\right) = -2x_4$

(1): $x_1 = -3x_2 + 2x_3 - x_5$

$= -3x_2 + 2(-2x_4) - x_5$

$= -3x_2 - 4x_4 - 2x_5$

Let $x_2 = r$, $x_4 = s$, $x_5 = t$:

$$\begin{cases} x_1 = -3r - 4s - 2t, \quad x_2 = r, \\ x_3 = -2s, \quad x_4 = s, \\ x_5 = t, \quad x_6 = 1/3, \quad r, s, t \in \mathbb{R}. \end{cases}$$

Gaussian Elimination

1. Produce a LE in the upper-leftmost corner.
2. Use the LE to clear the entries below.
3. Send zero rows to the bottom.
4. Repeat!

Matrices

A **matrix** is a rectangular array of numbers, called **entries**. The **size** of a matrix is described in terms of the number of rows and columns.

Ex. $\begin{pmatrix} 1 & 2 \\ 3 & 1 \\ -1 & 1 \end{pmatrix}_{3\times 2}$, $\begin{pmatrix} 2 & 1 & \sqrt{\pi} \end{pmatrix}_{1\times 3}$

More generally:

$$A_{m\times n} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

The (i,j)-entry is a_{ij}.

If $m=n$, then A is a **square** matrix.

Ex. Here are some important square matrices:

- diagonal matrix: $a_{ij} = 0$ when $i \neq j$

$$\begin{pmatrix} d_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & d_n \end{pmatrix}$$

- identity matrix: $a_{ij} = \begin{cases} 1 & \text{when } i=j, \\ 0 & \text{when } i\neq j. \end{cases}$

$$I_n = \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{pmatrix}$$

Operations on Matrices

Let A, B be two $m \times n$ matrices. Then:

- $A = B \iff a_{ij} = b_{ij}$ for all i, j.
- $A + B = (a_{ij} + b_{ij})_{m \times n}$
- $cA = (ca_{ij})_{m \times n}$ for $c \in \mathbb{R}$

New Operations on Matrices

Let A be an $m \times n$ matrix. The transpose of A is obtained by interchanging the rows and columns of A:

$$\text{if } A = (a_{ij}), \text{ then } A^T = (a_{ji})$$

If $A = A^T$, then A is symmetric:

$$a_{ij} = a_{ji} \text{ for all } i, j.$$

Matrix Multiplication

Let $A = (a_{ij})_{m \times p}$ and $B = (b_{ij})_{p \times n}$.
The matrix product AB is the $m \times n$ matrix whose entries are given by

$$(AB)_{ij} = \sum_{k=1}^{p} \underbrace{a_{ik}}_{\text{ith row of } A} \underbrace{b_{kj}}_{\text{jth col. of } B}.$$

Ex. $A = \begin{pmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{pmatrix}_{2 \times 3}$

$B = \begin{pmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{pmatrix}_{3 \times 4}$

The product AB is 2×4:

$$\begin{pmatrix} 12 & - & - & - \\ - & - & - & - \end{pmatrix}$$

$(AB)_{11} = 1 \cdot 4 + 2 \cdot 0 + 4 \cdot 2 = 12.$